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An Exposition of Gamma Distributions in Physics

Course Project Report

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1 Introduction

The impact of statistics theory on modern day science discoveries cannot be overstated. All the fancy artificial intelligence and machine learning techniques are rooted deep in statistical theory. It, therefore, becomes increasingly essential to ground ourselves as scientists in learning statistics correctly and ethically. The Normal Distribution (also known as the Gaussian) is perhaps the single most important probability distribution in studying scientific phenomena and social science experiments. Clearly, it is best candidate for this due to the Central Limit Property. However, an overlooked probability distribution that is of key importance is the Gamma Distribution. In my study, I aim to dive into the derivation of the Gamma distribution, study its properties that make it desirable in modeling, and uncover hidden connections between it and physical phenomena.

2 Derivation of Gamma Distribution

2.1 The Gamma Distribution

We studied the **Gamma Function**, which is defined by the Euler Integral [1] as:

$$\Gamma(z) = \int_0^{\infty} e^{-t} t^{z-1} dt, \quad \text{Re}(z) > 0$$

It is easy to see that we can build a probability distribution out of this function by dividing by the Gamma Function value from the integrand. That ensures that our probability function integrates into 1. We can change the variable from $x \rightarrow x/\beta$ to further generalize the distribution, making it in a more desirable form for future applications. Let $X \sim \Gamma(\alpha, \beta)$, then the 2-parameter **Gamma Distribution** for X is:

$$f_X(x) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}, & x > 0, \\ 0, & x \leq 0. \end{cases}$$

Our goal is to study this probability distribution and to determine whether it models real-world phenomena. We will start by stating a physical problem that requires a model to analyze. Then, derive from first probability principles the distribution to model it. This excursion will require us to delve into stochastic process such as the bernoulli and poisson processes. We will also explore many distributions from binomial, poisson, exponential and erlang. These distributions and processes are incredibly useful in scientific phenomena that evolve with time from a plethora of fields biology, chemistry, physics and the social sciences. So, it will be well worth it to learn them.

Problem Statement: How long does it take for the k th particle from a homogeneous source to arrive at a detector?

Obtaining a model for the time it takes for arbitrary k particles to reach a detector gives clues to the particle flux produced by the source and the underlying physical interactions to produce that flux. This type of experiment is best modeled by a **stochastic process**: A model of a probabilistic experiment that evolves in time and generates a sequence of random variables. In an effort to model our particle detection problem, we switch our focus to a process we are quite familiar with: modeling the number of heads flipped from a coin n times.

2.2 The Binomial Distribution

We begin by modeling a coin toss.

Definition 1

Let X be a random variable taking only 2 values, 0 or 1, with probability p , if $X = 1$. Then, X is Bernoulli distributed with parameter p :

$$p_X(x) = \begin{cases} p & \text{if } x = 1, \\ 1 - p & \text{if } x = 0. \end{cases}$$

with $E[X] = p$ and $Var(X) = p(1 - p)$.

If X describes a fair coin toss representing $X = 1$ as heads and $X = 0$ as tails. Then, $X \sim \text{Bernoulli}(\frac{1}{2})$.

To model the number of heads after n tries, we go into the territory of modeling a sum of independent and identically distributed Bernoulli variables X_i .

Definition 2

Let $X_i \sim \text{Bernoulli}(p)$. Then, $X_1 + X_2 + \dots + X_n$ is distributed according to a binomial distribution with parameters n (number of tries) and p (probability of success). The binomial probability distribution is:

$$p_X(x) = \begin{cases} \binom{n}{p} p^x (1 - p)^{n-x} & \text{for } 0 \leq x \leq n, \\ 0 & \text{otherwise.} \end{cases}$$

with $E[X] = np$ and $Var(X) = np(1 - p)$.

We can understand the binomial distribution further by solving an example.

Problem 3

What is the probability of getting 6 heads in 10 fair coin tosses?

Solution

The probability of getting one arrangement of 6 heads and 4 tails is: $p^6(1 - p)^4$.

There are $\binom{10}{6}$ different arrangements in which we get 6 heads from 10 tosses.

Therefore, $P(6 \text{ heads}) = \binom{10}{6} p^6 (1 - p)^4 = \binom{10}{6} (0.5)^{10}$

We can visualize the PMF of a $\text{Binomial}(10, p)$ through the figures below:

We can also think of the binomial distribution as a discrete stochastic process. Consider each coin toss X_i being recorded in specific time interval τ , for instance 1 second. Then, we have a successive collection of X_i for a time period of $T = \tau n$. This shifts the Binomial distribution into a process that evolves over time. This process is called a **Bernoulli process**.

The Bernoulli process can provide some insights into our particle detector problem. While the Bernoulli process is discrete, the arrival of our particles occur in continuous time. However, we can get around this problem by studying limiting behavior of the Bernoulli process as time intervals τ shrink to 0 considering infinite time intervals in time T .

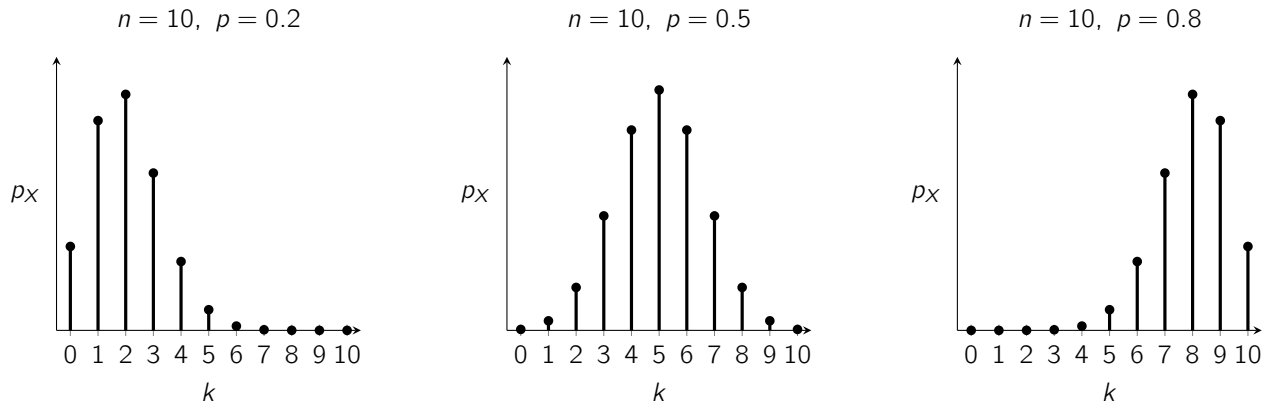


Figure 1: PMFs of three binomial distributions.

This is equivalent to making the number of tries n tend to ∞ , the probability of a success p shrink to 0 yet keeping the mean $np = \lambda$ in our time interval T constant. This limiting distribution is called the **Poisson distribution**. A graph of the PMF of a Binomial distribution with large n and small p is:

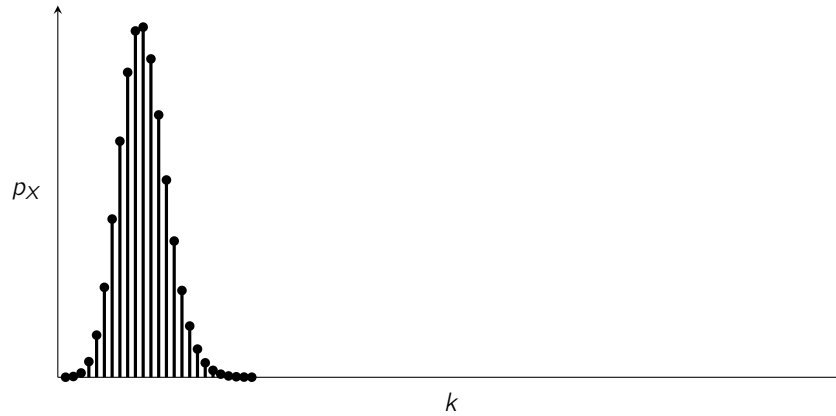


Figure 2: Binomial PMF with $n = 100$ and $p = 0.1$.

2.3 The Poisson Distribution

We proceed to derive the Poisson distribution by conducting the limiting behavior on the Binomial distribution.

Theorem 4

Let $X \sim \text{Binomial}(n, p)$. If $n \rightarrow \infty$, $p \rightarrow 0$ and $\lambda = np$ is kept constant, then X is Poisson distributed with parameter λ . Its PMF is:

$$p_X(x) = \begin{cases} \frac{\lambda^k}{k!} e^{-\lambda} & \text{for } k = 0, 1, 2, \dots, \\ 0 & \text{otherwise.} \end{cases}$$

with $E[X] = \lambda$ and $\text{Var}(X) = \lambda$.

Proof. Starting from the binomial PMF,

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k},$$

we impose the scaling $p = \lambda/n$, so that $np = \lambda$ remains fixed. Then

$$\mathbb{P}(X = k) = \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}.$$

For fixed k and $n \rightarrow \infty$, we use the limits

$$\binom{n}{k} \left(\frac{1}{n}\right)^k = \frac{n(n-1) \cdots (n-k+1)}{k! n^k} \rightarrow \frac{1}{k!},$$

and

$$\left(1 - \frac{\lambda}{n}\right)^n \rightarrow e^{-\lambda}, \quad \left(1 - \frac{\lambda}{n}\right)^{-k} \rightarrow 1.$$

Hence,

$$\mathbb{P}(X = k) \rightarrow \frac{\lambda^k}{k!} e^{-\lambda}.$$

This is exactly the PMF of the Poisson distribution with parameter λ . □

The Poisson distribution models the number of events occurring in a fixed time interval. If particles arrive randomly in time at a constant average rate λ , then the probability of observing k arrivals in a time interval of length τ is given by

$$P(k, \tau) = \mathbb{P}(k \text{ arrivals occur in time interval } \tau).$$

Thus, in this setting, the parameter λ has a natural interpretation: it is the arrival rate of particles per unit time.

To build the Poisson process, we impose three properties.

1. **Time-homogeneity.** The probability $P(k, \tau)$ depends only on the length τ of the interval, not on where the interval is located in time.
2. **Independent increments.** The number of arrivals in disjoint time intervals are independent.
3. **Small interval probabilities.** For sufficiently small τ ,

$$P(1, \tau) = \lambda\tau + O(\tau^2),$$

$$P(0, \tau) = 1 - \lambda\tau + O(\tau^2),$$

and, for $k \geq 2$,

$$P(k, \tau) = O(\tau^2).$$

The third condition is especially important. The Poisson model allows multiple arrivals in a time interval, but if the interval is very small, then the probability of two or more arrivals should be negligible compared to the probability of exactly one arrival. In other words, as $\tau \rightarrow 0$, the dominant possibilities are either no arrival or exactly one arrival. This makes the approximation

$$P(1, \tau) \approx \lambda\tau, \quad P(0, \tau) \approx 1 - \lambda\tau$$

intuitively reasonable.

These properties are enough to completely determine the process. In fact, they lead to

$$P(k, \tau) = \frac{(\lambda\tau)^k}{k!} e^{-\lambda\tau}, \quad k = 0, 1, 2, \dots$$

which is the Poisson distribution with mean $\lambda\tau$. Therefore, the number of arrivals in any interval of length τ is Poisson distributed:

$$N(\tau) \sim \text{Poisson}(\lambda\tau).$$

2.4 The Exponential Distribution

The Poisson process models the number of particles arriving in a time interval. However, our main interest is slightly different. Rather than counting how many arrivals occur by time t , we want to study the *time* we must wait until a given number of arrivals has occurred. To do this, we begin with the waiting time for the first arrival.

Theorem 5

Let $X_1, X_2, \dots, X_n$ be a Poisson process with parameter $\lambda\tau$. The time until the first arrival T_1 has an **exponential** distribution with PDF:

$$f_{T_1} = \lambda e^{-\lambda t}, \quad t \geq 0$$

$$\text{with } E[T_1] = \frac{1}{\lambda} \text{ and } \text{Var}(T_1) = \frac{1}{\lambda^2}$$

Proof. We proceed by trying to find the CDF F_{T_1} then differentiating to get the PDF f_{T_1} . Let T_1 denote the time until the first arrival. Then

$$\mathbb{P}(T_1 > t) = \mathbb{P}(\text{no arrivals in } [0, t]).$$

Since the number of arrivals in $[0, t]$ is Poisson with parameter λt ,

$$\mathbb{P}(T_1 > t) = P(0, t) = e^{-\lambda t}.$$

Therefore, the cumulative distribution function of T_1 is

$$F_{T_1}(t) = \mathbb{P}(T_1 \leq t) = 1 - e^{-\lambda t}, \quad t \geq 0.$$

Differentiating gives the density

$$f_{T_1}(t) = \lambda e^{-\lambda t}, \quad t \geq 0.$$

So the waiting time until the first arrival is exponentially distributed:

$$T_1 \sim \text{Exp}(\lambda).$$

□

2.5 The Erlang Distribution

We are now ready to consider the waiting time for k arrivals. Let T_k denote the time until the k th arrival. We may write

$$T_k = Y_1 + Y_2 + \cdots + Y_k,$$

where

$$Y_1 = T_1, \quad Y_2 = T_2 - T_1, \quad Y_3 = T_3 - T_2, \quad \dots$$

are the interarrival times.

A fundamental property of the Poisson process is that it has the *memoryless* or *fresh start* property: after each arrival, the process probabilistically starts over. As a consequence, the interarrival times $Y_1, Y_2, \dots$ are independent and identically distributed exponential random variables with parameter λ :

$$Y_i \sim \text{Exp}(\lambda), \quad i = 1, 2, \dots$$

Thus T_k is the sum of k independent exponential random variables with common rate λ . The density of this sum can be found by investigating the CDF of a sum of iid exponential variables.

Theorem 6

Let T_k denote the time for the k th arrival in a Poisson process. Then, T_k is **Erlang** distributed with parameters λ and k with PDF:

$$f_{T_k}(t) = \frac{\lambda^k t^{k-1} e^{-\lambda t}}{(k-1)!}, \quad t \geq 0.$$

$$\text{with } \mathbb{E}[T_k] = \frac{k}{\lambda}, \quad \text{Var}(T_k) = \frac{k}{\lambda^2}.$$

Proof. We proceed by a similar proof of the PDF of T_1 . Note that for a given time $t \geq 0$, the event $\{T_k \leq t\}$ is the same as the event that there have k arrivals or more by time t .

Thus the cumulative distribution function of T_k is

$$F_{T_k}(t) = \mathbb{P}(T_k \leq t) = \sum_{n=k}^{\infty} P(n, t) = 1 - \sum_{n=0}^{k-1} P(n, t) = 1 - \sum_{n=0}^{k-1} \frac{(\lambda t)^n e^{-\lambda t}}{n!}.$$

Differentiating, we obtain

$$f_{T_k}(t) = \frac{d}{dt} F_{T_k}(t) = -\frac{d}{dt} \left(\sum_{n=0}^{k-1} \frac{(\lambda t)^n e^{-\lambda t}}{n!} \right) = \lambda e^{-\lambda t} \frac{(\lambda t)^{k-1}}{(k-1)!}.$$

Therefore,

$$f_{T_k}(t) = \frac{\lambda^k t^{k-1} e^{-\lambda t}}{(k-1)!}, \quad t \geq 0.$$

□

If we look more closely, the Erlang distribution is just a special case of the Gamma distribution. Recall that if $X \sim \text{Gamma}(\alpha, \beta)$, with shape parameter α and scale parameter β , then its density is

$$f_X(x) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}, \quad x \geq 0.$$

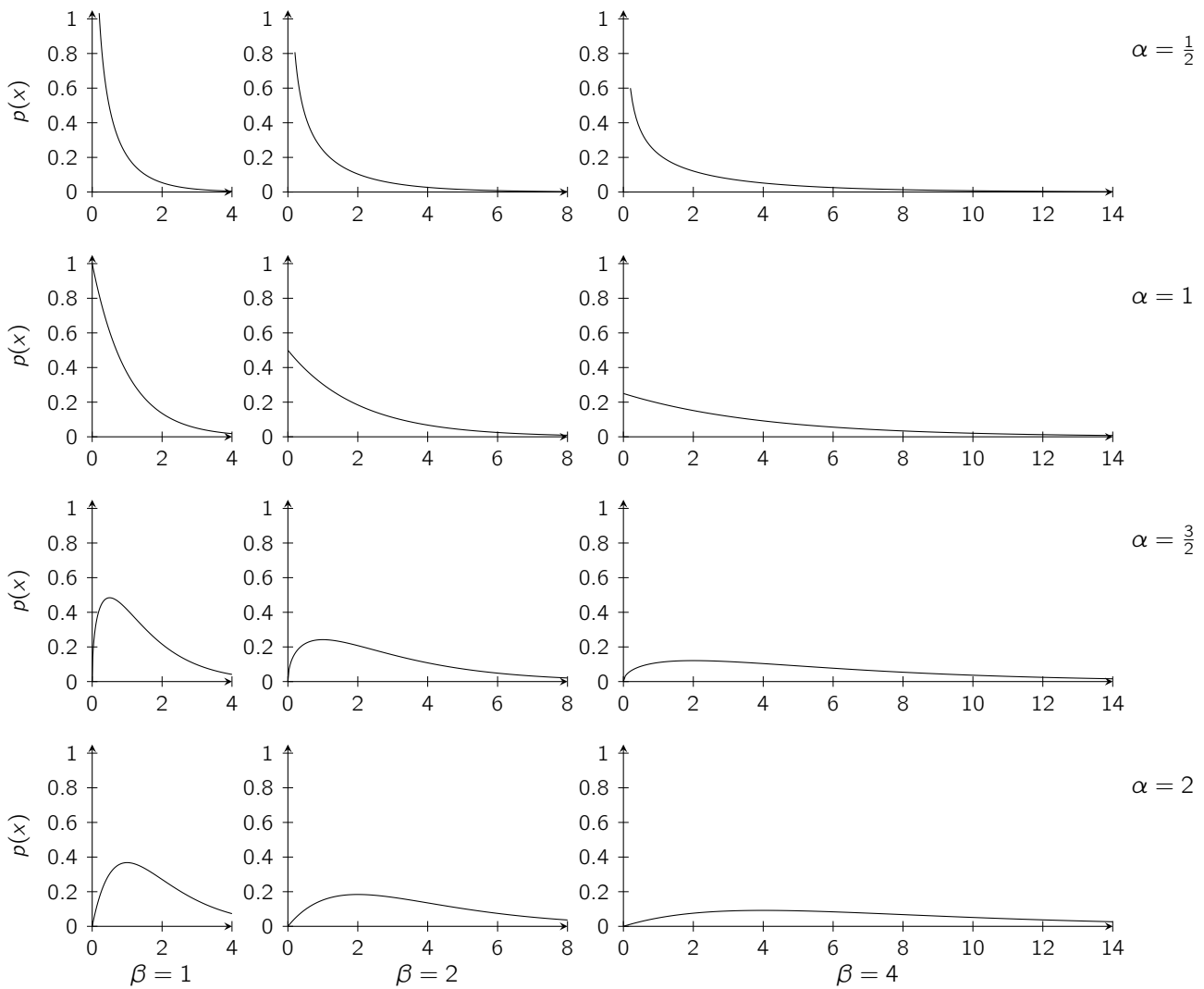
Comparing this with the density of T_k , we see that

$$T_k \sim \text{Gamma}(k, 1/\lambda).$$

So the Erlang distribution corresponds exactly to the Gamma distribution when the shape parameter is an integer $\alpha = k$ and the scale parameter is the reciprocal of the arrival rate $\beta = \frac{1}{\lambda}$. We finally have a statistical interpretation to what the Gamma distribution models. It models the waiting time until the k th arrival in a Poisson process.

2.6 Graphs of the Gamma Distribution

Graphs of probability distribution drive our intuition further with investigating their parameters and key features. Adapted from [2], we find some graphs of the Gamma distribution with different shape α and scale β parameters. Each row of graphs represent a special type of gamma distribution namely in order, χ^2 distribution with 1 df, exponential distribution, χ^2 distribution with 3 df and an Erlang distribution with $k = 2$.



2.7 Properties of Gamma distribution

The Gamma distribution has many properties that make it desirable. Here are some of the most important.

Fact 7 (Mean, variance, and mode of the Gamma distribution)

Let $X \sim \text{Gamma}(\alpha, \theta)$ with shape $\alpha > 0$ and scale $\theta > 0$, so that

$$f_X(x) = \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} e^{-x/\theta}, \quad x > 0.$$

Then

$$\mathbb{E}[X] = \alpha\theta, \quad \text{Var}[X] = \alpha\theta^2, \quad \text{mode}(X) = (\alpha - 1)\theta \quad \text{for } \alpha \geq 1.$$

Fact 8 (Scaling property of the Gamma distribution)

Let $X \sim \text{Gamma}(\alpha, \theta)$ and let $c > 0$. Then

$$cX \sim \text{Gamma}(\alpha, c\theta).$$

Fact 9 (Sum of independent Gamma random variables)

Let $X_1, \dots, X_n$ be independent random variables with

$$X_i \sim \text{Gamma}(\alpha_i, \theta),$$

all having the same scale parameter θ . Then

$$X_1 + \dots + X_n \sim \text{Gamma}(\alpha_1 + \dots + \alpha_n, \theta).$$

We interpret the scaling and summing properties using Poisson processes. If we scale timings T_k by a constant c , then the arrival rate parameter $1/\theta$ will be reduced by a factor $1/c$. If we add timings T_k with different k values, then we get a Poisson process with the same arrival rate but k_i arrival positions summed up from each Poisson process.

Fact 10 (Sum of squares of Gaussians[3])

Let $Z_1, \dots, Z_k$ be independent standard normal random variables, $Z_i \sim \mathcal{N}(0, 1)$, and define

$$Q = \sum_{i=1}^k Z_i^2.$$

Then

$$Q \sim \chi_k^2,$$

and equivalently

$$Q \sim \text{Gamma}\left(\frac{k}{2}, 2\right).$$

χ^2 distributions are well known in statistics for being the distribution of the scaled sample variance, but they are also known as the Boltzmann distribution for ideal gases in statistical mechanics.

Fact 11 (Gamma kernel as the Laguerre weight [1])

The associated Laguerre polynomials $L_n^{(\alpha)}(x)$, with $\alpha > -1$, are orthogonal on $(0, \infty)$ with respect to the weight

$$w_\alpha(x) = x^\alpha e^{-x},$$

that is,

$$\int_0^\infty x^\alpha e^{-x} L_m^{(\alpha)}(x) L_n^{(\alpha)}(x) dx = 0, \quad m \neq n.$$

Since

$$x^\alpha e^{-x} = x^{(\alpha+1)-1} e^{-x},$$

this weight is, up to normalization by $\Gamma(\alpha + 1)$, the kernel of a Gamma density with shape parameter $\alpha + 1$ and scale 1.

This was quite surprising to me that there was another interpretation to the Gamma distribution that is useful in solving associated Laguerre ODEs. These show up in the radial electron density of the hydrogen atom in quantum mechanics.

Fact 12 (Maximum entropy characterization of the Gamma distribution [4])

Among all probability densities f on $(0, \infty)$ satisfying

$$\int_0^\infty f(x) dx = 1, \quad \int_0^\infty x f(x) dx = E[X], \quad \int_0^\infty \ln x f(x) dx = E[\ln X],$$

the unique density maximizing the differential entropy

$$H(f) = - \int_0^\infty f(x) \ln f(x) dx$$

is the Gamma density

$$f(x) = \frac{1}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} e^{-x/\theta}, \quad x > 0,$$

The entropy defined in the statistical sense here is analogous to the entropy in the statistical mechanics sense. This is also another reason why we see the gamma distribution popping up for energy distributions. It maximizes the entropy!

3 Applications of the Gamma Distribution

3.1 Astroparticle Physics

Problem: How can we reconstruct the direction of a neutrino coming from galactic & extragalactic sources? [5]

3.1.1 Tracking Neutrino through Muons

Modern neutrino telescopes such as Icecube and KM3NeT do not detect neutrinos directly. Instead, they detect the charged particles that a neutrino leaves behind after interacting with matter, kind of like seeing footprints rather than the particle itself. Neutrinos come in three flavours, ν_e , ν_μ , and ν_τ , and in a charged-current interaction with a nucleon N they produce the corresponding charged lepton:

$$\nu_\ell + N \rightarrow \ell + X, \quad \ell \in \{e, \mu, \tau\},$$

where X are residual particles produced.

Among these leptons, muons are especially useful for neutrino astronomy. Electrons usually interact with water molecules producing a shower of photons that are hard to retrace their direction. Taus decay very quickly after being produced. However, muons leave long, approximately straight tracks, which makes them much better for reconstructing the direction of the parent neutrino.

Most of the muons entering a detector from above are not the signal we are after. They are atmospheric muons produced when cosmic rays strike the Earth's atmosphere mostly sourced from the sun. However, sources of greatest interest are those from galactic or extragalactic sources. A key idea is that the Earth acts as a filter: muons cannot pass through the whole Earth, but neutrinos can. An astrophysical neutrino can travel through the Earth, interact near the detector, and produce an up-going muon. By reconstructing the track and energy of that muon, we can infer information about the direction of the original neutrino and about the source that may have produced it.

When a relativistic muon moves through ice or water faster than the phase velocity of light in that medium, it emits **Cherenkov radiation**, which is equivalent to a sonic boom for sound waves. This does not mean the muon is moving faster than light in vacuum; it only means that it is moving faster than light propagates in the medium.

This Cherenkov light is recorded by **photomultiplier tubes** (PMTs), which are very sensitive light sensors that convert a small flash of light into an electrical pulse. For each PMT hit, the detector records a hit time, which we denote by t_{hit} . To reconstruct a muon track, one starts with a trial track characterized by a point $\mathbf{r}_0$ on the track, a time t_0 at which the muon passes that point, and a direction $\hat{\mathbf{p}}$. If a PMT is located at position $\mathbf{r}_i$, then the expected arrival time of an ideal direct Cherenkov photon is

$$t_{\text{geo},i} = t_0 + \frac{\hat{\mathbf{p}} \cdot (\mathbf{r}_i - \mathbf{r}_0) + d_i \tan \theta_c}{c_{\text{vac}}},$$

where d_i is the perpendicular distance from the PMT to the track and c_{vac} is the speed of light in vacuum.

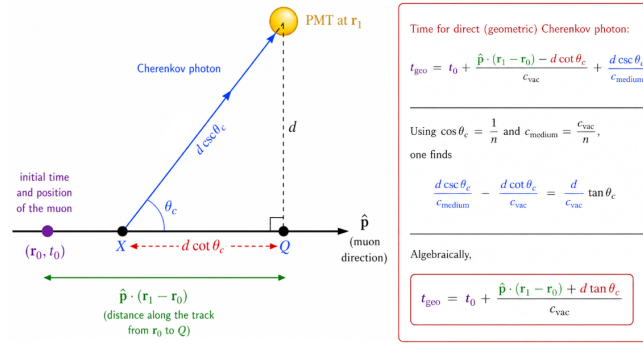


Figure 3: Calculation of t_{geo} for detecting cherenkov light

The central random quantity is then the **time residual**

$$t_{\text{res},i} = t_{\text{hit},i} - t_{\text{geo},i}.$$

3.1.2 Modeling Time Residuals

If the medium were perfectly transparent and the detector ideal, then one would expect $t_{\text{res}} = 0$. In reality, however, the photons are delayed by several effects. The dominant one is scattering in the ice or water: instead of taking the shortest path to the PMT, the photon may scatter one or more times, increasing its path length and hence its arrival time. There is also detector jitter, which is the uncertainty in the timing electronics and PMT response. In addition, there is noise, such as dark counts or electronic artefacts, where a PMT records a hit even though no relevant signal photon arrived. Finally, there can be extra Cherenkov light from secondary showers along the muon track, for example from stochastic radiative energy losses, and these photons often arrive later than the ideal direct light.

In neutrino-telescope reconstruction, the time-residual t_{res} distribution is often modelled by a gamma-like function called the **Pandel function**:

$$p(t_{\text{res}}) = \frac{1}{N(d)} \frac{\tau^{-d/\lambda}}{\Gamma(d/\lambda)} t_{\text{res}}^{d/\lambda-1} \exp\left[-t_{\text{res}} \left(\frac{1}{\tau} + \frac{c_{\text{medium}}}{\lambda_a}\right) - \frac{d}{\lambda_a}\right], \quad t_{\text{res}} > 0,$$

with normalization

$$N(d) = e^{-d/\lambda_a} \left(1 + \tau \frac{c_{\text{medium}}}{\lambda_a}\right)^{-d/\lambda}.$$

Here, d is the distance from the muon track to the PMT, λ is a characteristic length scale controlling the shape of the distribution, τ is a characteristic time scale, λ_a is the absorption length of the medium, and $c_{\text{medium}} = \frac{c_{\text{vac}}}{n}$ is the speed of light in the medium. The core of the Pandel function has the same structure as a gamma distribution,

$$p(t_{\text{res}}) \propto t_{\text{res}}^{k-1} e^{-\beta t_{\text{res}}},$$

with parameters

$$k = \frac{d}{\lambda}, \quad \beta = \frac{1}{\tau} + \frac{c_{\text{medium}}}{\lambda_a}.$$

So, the Pandel function is gamma-like, with additional distance-dependent factors that account for absorption and detector geometry.

3.1.3 Statistical Interpretation

There is also a useful statistical intuition for why a gamma distribution appears here. The total delay can be thought of as the sum of many small positive delay contributions:

$$t_{\text{res}} \approx \Delta t_1 + \Delta t_2 + \cdots + \Delta t_N.$$

The farther the PMT is from the muon track, the more likely it is that the photon has undergone multiple scattering events, so the effective number of delay contributions increases with distance. If these individual contributions are approximately exponential,

$$\Delta t_i \stackrel{\text{iid}}{\sim} \text{Exp}(\beta),$$

then their sum is gamma distributed:

$$\sum_{i=1}^N \Delta t_i \sim \text{Gamma}(N, \beta).$$

Once the pdf for t_{res} is specified, the track can be reconstructed with a maximum-likelihood method. The reconstruction asks: which hypothetical muon track makes the observed pattern of PMT hit times t_{res} most probable? The best-fit track with initial position $\mathbf{r}_0$, initial time t_0 , and momenta $\hat{\mathbf{p}}$ is the one that maximizes this likelihood. We can finally gain information on the source of neutrino produced in the night sky from the reconstructed track.

3.2 Statistical Mechanics

Problem: What is the energy distribution of a 3D ideal gas? [6]

We consider a classical ideal gas in three dimensions at temperature T , with

$$\beta = \frac{1}{k_B T}.$$

Since the gas is ideal, the particles do not interact, and the Hamiltonian for a single particle is purely kinetic:

$$E = \frac{p_x^2 + p_y^2 + p_z^2}{2m}.$$

Our goal is to derive the probability distribution of the single-particle energy E , and then compute the mean energy and the energy fluctuations. We present two complementary derivations: one from statistical mechanics and one from probability theory.

3.2.1 Statistical Mechanics Derivation

In the canonical ensemble, the probability density of a microstate is proportional to the **Boltzmann factor**,

$$P(s) \propto e^{-\beta E(s)}.$$

To obtain the energy distribution, we must also include the density of states.

For a single particle in three dimensions, the number of momentum states with momentum in the shell between p and $p + dp$ is proportional to the volume element in momentum space,

$$d^3p = 4\pi p^2 dp.$$

Since

$$E = \frac{p^2}{2m}, \quad \text{so that} \quad p = \sqrt{2mE},$$

we have

$$dp = \frac{m}{p} dE.$$

Therefore,

$$d^3p \propto p^2 dp \propto p^2 \frac{m}{p} dE \propto p dE \propto \sqrt{E} dE.$$

Hence the density of states as a function of energy is

$$g(E) \propto E^{1/2}.$$

The probability density for the energy is therefore

$$P(E) dE \propto g(E) e^{-\beta E} dE \propto E^{1/2} e^{-\beta E} dE.$$

Normalizing, we get the Gamma distribution:

$$P(E) = \frac{\beta^{3/2}}{\Gamma(3/2)} E^{1/2} e^{-\beta E}, \quad E \geq 0.$$

with parameter $\alpha = \frac{3}{2}$ and $\theta = \frac{1}{\beta} = k_B T$.

3.2.2 Probabilistic Derivation

The same result can be obtained directly from probability theory.

For a particle in thermal equilibrium, each momentum component is Gaussian:

$$p_x, p_y, p_z \sim \mathcal{N}(0, mk_B T),$$

independently. Define the normalized variables

$$X_i = \frac{p_i}{\sqrt{mk_B T}}, \quad i = 1, 2, 3.$$

Then each $X_i \sim \mathcal{N}(0, 1)$, so

$$X_1^2 + X_2^2 + X_3^2 \sim \chi_3^2.$$

Now the kinetic energy is

$$E = \frac{p_x^2 + p_y^2 + p_z^2}{2m} = \frac{k_B T}{2} (X_1^2 + X_2^2 + X_3^2).$$

Thus

$$\frac{2E}{k_B T} \sim \chi_3^2 \sim \text{Gamma}\left(\frac{3}{2}, 2\right).$$

Thus, scaling with $\frac{1}{2\beta}$ yields our PDF:

$$P(E) = \frac{\beta^{3/2}}{\Gamma(3/2)} E^{1/2} e^{-\beta E}, \quad E \geq 0.$$

Since this is a Gamma distribution with shape $\alpha = 3/2$ and scale $\theta = 1/\beta$, its moments follow immediately:

$$\langle E \rangle = \alpha \theta = \frac{3}{2} k_B T,$$

$$\text{Var}(E) = \alpha \theta^2 = \frac{3}{2} (k_B T)^2.$$

Thus, the energy distribution of a 3D ideal gas follows the Gamma distribution, and the result is fully consistent with the equipartition theorem.

3.3 Quantum Mechanics

Problem: What is the probability distribution resulting from the radial wave function of an electron in a hydrogen atom? [1]

3.3.1 Solving the Schrödinger Equation

Another way to think about the Gamma distribution, besides the statistical mechanics picture of sums of squares of Gaussians and the astroparticle physics picture of inter-arrival times, is through its connection to the associated Laguerre functions. These arise in the radial part of the Schrödinger equation for an electron in a hydrogen atom.

The time-independent Schrödinger equation for a hydrogen-like atom is

$$-\frac{\hbar^2}{2m} \nabla^2 \psi - \frac{Ze^2}{4\pi\epsilon_0 r} \psi = E\psi.$$

Because the Coulomb potential is spherically symmetric, we separate variables and find the radial function then satisfies

$$-\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \frac{Ze^2}{4\pi\epsilon_0 r} R + \frac{\hbar^2}{2m} \frac{\ell(\ell+1)}{r^2} R = ER.$$

Introducing a dimensionless radial variable

$$\rho = \alpha r, \quad \alpha = \frac{2Z}{na_0},$$

with a_0 the Bohr radius, the bound-state solutions take the form

$$R_{n\ell}(r) \sim e^{-\rho/2} \rho^\ell \times (\text{associated Laguerre polynomial}).$$

3.3.2 The Radial Probability Distribution

Squaring the wavefunction and including the radial measure $r^2 dr$, the radial probability density becomes

$$P_{n\ell}(r) dr = |R_{n\ell}(r)|^2 r^2 dr.$$

In terms of ρ , this has the form

$$P_{n\ell}(\rho) d\rho \propto e^{-\rho} \rho^{2\ell+2} \left[L_{n-\ell-1}^{(2\ell+1)}(\rho) \right]^2 d\rho.$$

Notice that the factor

$$e^{-\rho} \rho^{2\ell+1}$$

is precisely of the same form as the kernel of a Gamma distribution,

$$x^{k-1} e^{-x},$$

up to normalization. In other words, the Gamma kernel appears naturally as the *weight function* for the associated

Laguerre polynomials. We find that the associated Laguerre polynomials satisfy the orthogonality relation

$$\int_0^\infty x^k e^{-x} L_m^k(x) L_n^k(x) dx = \frac{(n+k)!}{n!} \delta_{mn}.$$

Thus the connection to the Gamma distribution here is not primarily statistical. It is instead analytic: the Gamma kernel is the natural weight function with respect to which the associated Laguerre polynomials are orthogonal, and this is exactly the structure that appears in the hydrogen radial equation.

3.3.3 Physical Interpretation

There is also a physical interpretation for why this form makes sense. The exponential factor

$$e^{-\rho/2}$$

ensures decay as $r \rightarrow \infty$, which is required for square-integrability of a bound state. Near the nucleus, the centrifugal term dominates, and regularity forces a power-law behavior

$$R_{n\ell}(r) \sim r^\ell.$$

So the overall radial solution must have the form

$$R_{n\ell}(r) \sim e^{-\rho/2} \rho^\ell \times \text{polynomial}.$$

The associated Laguerre polynomial provides this remaining polynomial factor, introducing the nodal structure of the excited states. However, the ground state, the probability distribution is indeed a Gamma!

4 Conclusion

Statistical models serve as a backbone to modeling uncertainty in many areas of science including physics. through our report, we have learned about the statistical derivation of the gamma distribution, its properties, and a couple of its applications in astroparticle physics, statistical mechanics, and quantum mechanics. As physicists, it is an incredibly exciting era to learn about statistical models from parametric, non-parametric and machine learning to devise novel models in explaining the workings of the world.

The statistical interpretation of the Gamma distribution is adapted from Tsitsiklis [7]. Many of the graphs were generated by code produced from AI. The graphs have been reviewed and verified by me personally.

References

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